

ARNAV JINDAL

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EDUCATION

Duke University

Aug. 2022 – May 2026

B.S. in Computer Science (Software Systems) & Economics – GPA: 3.6

Durham, NC

- **Relevant Coursework:** Computer Architecture, Operating Systems, Compilers, Databases, Data Structures, Analysis of Algorithms, Computer Graphics, Functional Programming, Audio & Image ML, Discrete Mathematics, Probability, Linear Algebra, Multivariable Calculus, Microeconomics, Macroeconomics, Econometrics, Asset Pricing & Risk

EXPERIENCE

Polymarket

Jan. 2026 – Jun. 2026

Platform Engineering

New York City, NY

- Architected a high-throughput data pipeline and warehouse, streaming hundreds of millions of trade records from execution engines and on-chain indexers into ClickHouse hot storage; owned infrastructure and end-to-end deployment.
- Partnered with the Polygon team to design and stress-test a new on-chain execution system, validating throughput against 100x projected traffic.
- Drove full-stack caching, database optimization, and schema migrations across a platform serving billions of requests; shipped real-time Bitcoin pricing with cached frontend hydration for low-latency market data.

Duke University, Dept. of Computer Science

Aug. 2025 – May 2026

Distinction Thesis Research – Advisor: Prof. Matthew Lentz

Durham, NC

- Shrank a Builder TEE's trusted computing base in Ethereum Proposer-Builder Separation (PBS/MEV), replacing full-node block simulation (Reth ~300K LoC, Lighthouse, 200GB+ state DB) with a thin stateless layer: 3 Reth crates + k256 validating blocks via execution-witness MPT proofs, in-memory state reconstruction, and REVM re-execution.
- Built a minimal Rust validation server, integrated it into the DPaaS/TEE-PBS auction, and upstreamed two Reth `debug_executionWitness` bug fixes; benchmarked 204 mainnet fixtures (77–272 ms) and end-to-end on 10,001 relay bids / 71 builders (~1.9 ms BLS verify, <1 ms winner selection).

IMC Trading

Jun. 2025 – Aug. 2025

Software Engineering Intern – D1 Execution

Chicago, IL

- Built a distributed Java Apache Flink cluster to statefully process 100M+ messages/day via Kafka Streams on Kubernetes; replaced a failing service with 60% lower latency, 2x throughput, and 100% accuracy.
- Analyzed C++ engine code and feed-data packets to improve MASS_CANCEL handling, order reductions, and order-lifecycle tracking on trading engines; collaborated with traders to fix strategies.

Alleviate Health

Aug. 2024 – May 2025

Founding Engineer

Remote

- Engineered agentic LLM voice/SMS workflows with tool/function calling and RAG over patient records to autonomously screen patients and book appointments.
- Built full-stack operational dashboards and backend orchestration; scaled the platform to 200K+ texts and 100K+ patient interactions.

Amazon, Inc.

May 2024 – Aug. 2024

Software Engineering Intern

Seattle, WA

- Built an internal CLI, distributed via Amazon's package manager, that triggered automated accessibility UI scans and cut manual testing time by 40%.
- Optimized scanner Lambdas through WebSocket proxies (50% faster scans, 25% lower cost) and shipped blue-green CI/CD pipelines with integrated alarms and metrics for production observability.

PROJECTS

- **Contract Exchange – Click Trading Simulator:** Distributed backend exchange with an order book, position tracking, and leveraged trades. Enforced price–time priority via RabbitMQ; SQLite + Redis for reliability. WebSocket trading UI used by 100+ teams in Duke's trading competition.
- **mRPC – Managed RPC Framework:** A gRPC alternative built with Dr. Matthew Lentz & Dr. Danyang Zhuo using centralized Rust proxies and improved marshaling for 100% faster RPC calls; added a PyO3/Rust-FFI Python frontend and co-authored the research paper.
- **Dagpi – Full-Stack SaaS Platform:** Next.js + TypeScript/Express.js app with a Python/FastAPI image microservice and Actix (Rust) auth/metrics services. Docker/Kubernetes + Nginx, Grafana/Prometheus, PostgreSQL/TimescaleDB, AWS Glacier backups, Stripe + PayPal gateways.
- **Polaroid – Python Image Library:** Integrated Rust via PyO3 FFI for faster filesystem access, I/O, and pixel-matrix computation. CI pipeline builds cross-platform wheels with Maturin and auto-generates documentation.
- **Graphics Engine – C++ CPU Rendering:** 2D engine with pixel-level rendering, primitives, blending, clipping, Bezier curves, matrix transforms, shader tiling, and fragment shaders; tuned via profiling, template metaprogramming, loop unrolling, and lookup tables.

TECHNICAL SKILLS

- **Languages:** Rust, C++, Python, Java, TypeScript, JavaScript, SQL, Solidity, Deno, Bash, MIPS
- **AI/LLM:** LangChain, RAG, vector stores (pgvector/Pinecone), agentic workflows, tool/function calling, MCP
- **Frameworks/Libraries:** Node.js, React, FastAPI, Flask, Actix, Apache Flink, Kafka, ClickHouse, PostgreSQL, Redis, RabbitMQ, Jupyter, Bazel, perf, Valgrind
- **Tools/Platforms:** Git, AWS, Docker, Kubernetes, Cloudflare, Vercel, Grafana, Prometheus, RPC